HACKATHON IN A BOX PLAYBOOK

A STEP-BY-STEP GUIDE FOR HOSTING A HACKATHON USING MATLAB AND SIMULINK

Reach out to hackathon@mathworks.com for questions

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HACKATHON-IN-A-BOX INTRODUCTION

OVERVIEW

Hackathons are events where students, hobbyists, and programmers of all levels get together for a short period of time and try to create something using software, hardware, and other materials. This Hackathon-in-a-Box will provide you with the steps and materials to host your own 4-6 hour hackathon in a 7 week timespan, though the length of the event and timeframe can be adjusted as needed.

TIMELINE

For the organizers, the competition will be divided into 4 phases over 7 weeks as shown below:

- *Phase 1: Event Planning*
  You make logistical decisions about the event and nominate the hackathon
- *Phase 2: Event Preparation*
  You prepare all materials for the competitors.
- *Phase 3: Open Registration and Promote*
  Students may begin registering for the event, and you will promote the event
- *Phase 4: The Competition*
  On the day of the event, you will introduce the competition and judging rubric, participants will develop and submit solutions, and judges will evaluate each submission so that winners may be announced.

This document focuses in detail on every phase that the competition will go through.

ORGANIZING TEAM STRUCTURE

The organizing team (you!) is responsible for organizing and promoting the event. This can be one person or several people, but responsibilities can generally be broken into the following roles:

- **Admin**: Responsible for majority of event pre-work, such as setting dates, booking the room(s), finalizing event-day resources, and coordinating all organizational efforts
- **Promotion**: Responsible for posters, social media, emails, and any other promotional activities
- **Technical Lead**: Responsible for choosing and understanding a problem statement and answering basic questions during the event.

We encourage you to work with clubs, student society chapters, or MathWorks Student Ambassadors if the event is at a school that has one.

HIGH LEVEL ROLES AND RESPONSIBILITIES

The roles and responsibilities of the organizing team are explained in detail throughout this document. At a high level, these responsibilities are as outlined in the table below.

Task	Timing	Phase
Identify Target Audience and how you will reach them	7 weeks prior to event	1
Choose a problem statement	7 weeks prior	1
Select a competition date	7 weeks prior	1
Select and reserve a venue	4+ weeks prior	1
Create Devpost page	5 weeks prior	2
(Optional) Set up workshop	4+ weeks prior	
Reach out to us for resources	5 weeks prior	2
Develop and execute a promotional plan	3 weeks prior	2
Fill in welcome slide deck	Week Of the event	2
Create scoring spreadsheet	Week Of the event	2
Print the certificates	Week Of the event	2
Open and promote registration	2-4 weeks prior	3
Send reminder update	The day before	3
Set up event space and welcome slides	The day of	4
Present welcome slides	The day of	4
Judge entries	The day of	4
Determine rankings and hand out prizes to first, second, and third place winners	The day of	4

PHASE 1: COMPETITION PLANNING

This phase will focus on the **pre-work to plan the competition**. It is recommended that process starts 7 weeks prior to the event date.

1.1. DECIDING LOGISTICS

1.1.1 IDENTIFYING THE TARGET AUDIENCE

- Identify the group(s) you will be marketing the event to.
    - For example, are you targeting a specific geographic region, school, club, or organization?
- Determine approximate attendance numbers. How many people are expected?
    - What is the maximum number of participants that is expected/will be permitted?
        - Keep in mind, more participants means you will need more staffing. It is okay to limit the number of participants allowed.
- Consider how you will be able to reach this group for promotion

- Is there an email list?
- Are there any social media channels or groups that can be utilized?
- Do you have any contacts within this group?

1.1.2 PICKING A PROBLEM STATEMENT

- Refer to the Problem Statements section for a list of options and the resources that come with each.
- Select **one** problem statement that participants will tackle during the event. Familiarize yourself with it.

1.1.3 DECIDING THE COMPETITION DATES

- Decide the Date, Start Time, and End Time (this should be 4-6 hours after the Start Time) of the hackathon

1.1.4 DECIDING THE VENUE

- The venue needs:
  - Enough room to safely host the maximum number of participants expected
  - Plenty of chairs, desk/table spaces, and accessible **outlets** for participants to use
    - You may wish to bring surge protectors to increase the availability of outlets
  - Free Wi-Fi access
  - A means of projecting a slide show from a laptop
- Once a venue has been chosen, reserve the space as soon as possible. While exact timelines will vary by venue type, this should typically be done at least 4 weeks in advance of the event.
  - Reserve the space for at least an hour before the event begins and an hour after the event concludes for set-up and clean-up.

PHASE 2: COMPETITION PRE-WORK

This phase will focus on the **pre-work to organize the competition**. It is recommended that process starts 6 weeks prior to the event date.

2.1. RESOURCE SETUP

2.1.1 CREATING THE DEVPOST EVENT

- Create a Devpost event where students can register for the Hackathon and submit final projects. This will also allow you to send them reminders and additional information as the event approaches.
- The steps for doing this are:
  - Go to https://devpost.com/
  - Log in or sign up for an account
  - Click on “Host a hackathon”, then “Host public hackathons”
  - Scroll to the bottom of the page and select “Start in-person hackathon”
  - In the form that opens, fill in the Hackathon name and event date. Then save the draft.
  - You will be brought to a dashboard for your new hackathon. Go through each sub-tab of the ‘Edit hackathon’ tab (Essentials, Eligibility, Design, Hackathon Site, etc.) and fill in the information.
  - On the Starter Kit Tab:

        - You can find a template registration email in the Communications Templates section that can help you fill in these sections.
    - On the 'Submissions' Tab:
        - The deadline is typically 45 minutes before the end of the event, unless you want more time for judging.
        - In the "Final Reminders" field, remind participants that their repository link needs to be public
        - Add Custom Submission Form questions:
            - Team Name
            - Team Members
            - Repository Link to Public Submission
            - What programming languages were used in your submission?
            - Are there any additional instructions required to view your demo?
    - Set Public Voting to off
    - On the Judging Tab:
        - Add at least one judge to the 'Judging' tab
        - Add the judging criteria categories and descriptions to the 'Criteria' field. Judging criteria can be found in the 'JUDGING AND SCORING' section of each Problem statement.
    - On the Prizes Tab:
        - Set the 'Winners Announced' field to be the end date and time of the event
        - For the 'Prizes' field:
            - If you are providing your own prizes, fill in this section based on the prizes you plan to provide.
            - If you would like to use MathWorks giveaways as prizes, reach out to us.
    - You are responsible for creating a code of conduct for this event, if required. We provide an example in the Code of Conduct document.
- Once all the fields are filled in, publish the page!

2.1.2 (OPTIONAL) SET UP WORKSHOP

- Decide if your technical lead will offer some kind of workshop or similar event leading up to the hackathon. Event types other than a workshop may include:
    - An onramp party – everyone gets together to complete a 1-2 hour onramp course, maybe with snacks
    - A Q&A session about the hackathon
- We do not provide materials for a workshop, but we have suggested onramps for each problem statement.
- This event should be 1-3 weeks prior to the hackathon and should not require that you are registered for the hackathon.
- Promote the hackathon at some point during this event.

2.1.3 REACH OUT TO US FOR RESOURCES

- We may be able to provide complimentary MATLAB access or provide prizes for your event if you reach out to us at hackathon@mathworks.com

    - Note: If the Hackathon is for a campus with a Campus-Wide offering of MATLAB, you may not need a license for the hackathon. You can check if your school has a Campus-Wide offering here
- This should be done at least 5 weeks prior to the event.

2.1.4 FILL IN WELCOME SLIDE DECK

- When students first arrive at the event, you should present a few slides that introduce participants to the problem and explain the schedule for the day.
- A template slide deck is provided here: Hackathon Introduction Slide Deck
    - Fill this in before the day of the event

2.1.5 CREATE SCORING SPREADSHEET

- Create an online spreadsheet that the team of judges will use to keep track of teach team's score during the event.
    - This spreadsheet should have four columns:
        - Team Name
        - Repository Link
        - Score
        - Notes
    - We recommend freezing the top row of this sheet. This can typically be done from a 'View' tab.
- Share this spreadsheet with each judge prior to the event.

2.1.6. CERTIFICATE PRINTING

- Edit the Winner Templates to reflect location and dates of competition.
- Print 5 of each of the first, second, and third place certificates. This will allow three teams of four to receive winner certificates, plus a few extras in case they are needed.

2.2. COMPETITION PROMOTION

2.2.1 DEVELOP PROMOTIONAL PLAN

- Local promotion should begin 3-4 weeks before the event.
- The Promotions Playbook lists several ideas for promotion.
- Start by deciding which promotional activities you will conduct, then determine when you will do each activity. Some common questions that the team will need to answer include:
    - What social media platforms will be used?
        - What channels will post?
        - When will social media posts go out?
    - When will promotional emails be sent?
    - When will reminder emails be sent to registrants?
    - Are there places we could hang or disburse posters and flyers?
    - Are there other events or meetings that we could use to promote the event?
- If applicable, consider working with the school or organization's IT department to send out promotional emails to the appropriate email lists.

- DO NOT reveal what the problem statement is, as this could give students the opportunity to begin coding before the event. You can provide some vague idea of what skills they may use however, such as:
  - "Learn how to collect sensor data from your phone and use it!"
  - "An opportunity to put your machine learning skills to the test!"
- You can utilize the poster and social media templates provided in this repository.

PHASE 3: OPEN REGISTRATION & PROMOTE

Once you have completed the pre-work, you can begin promoting the event and allow students to register.

3.1. REGISTRATION

3.1.1. SEND OUT REGISTRATION FORM

- At least 2 weeks before the event, send out an email and (if applicable) social media post(s) with the link to the Devpost page made previously.
- Participants may have formed a team before registering, but this is not required. They will have the opportunity to do this in the weeks leading up to and at the event.
  - Teams should have a minimum of one person and a maximum of four people.
- Participation should be free

3.2. COMMUNICATION WITH TEAMS

3.2.1 REMINDER UPDATE

- The day before the event, send a reminder update using the 'Updates' tab of Devpost. This update should include reminders about date, time, location, and should remind participants to bring a computer.

3.2.2 GETTING STARTED RESOURCES

- The day before the event (or earlier), add an announcement using the "Updates" tab on Devpost.
- This update should include an overview of the problem statement and the GitHub link to the student resources provided with the chosen problem statement.
- Choose to "Send and publish" "At a specific date and time" then set the date and time to be when the hackathon starts. That way all participants will get an email with the starting materials and instructions.

3.3 STAFFING THE EVENT

As participants register for the event, you will get a sense of how many participants will be attending the event.

Initial staffing numbers will be based on this estimated participation. It is better to get too many judges than too few. You can always reduce staffing as is appropriate.

3.3.1 WELCOME STAFF

- Regardless of how many students register, staff at least two individuals as the Welcome Staff.
- One person will be responsible for setting up and manning the welcome table.
- One person will be responsible for setting up and delivering the welcome presentation.

3.3.2 JUDGES

- We recommend one Judge per 15 participants. They only need to be at the final hour of the event.
    - Judges can also be Welcome Staff
- Consider working with student organizations or graduate students if you do not have enough people to judge submissions.
    - Allow them to familiarize themselves with the problem statement and potential solutions ahead of time.

3.3.3 GUEST SPEAKERS

- If you know of any other individuals relevant to the hackathon location/school/organization that it may be beneficial to include (such as department chairs), you may wish to invite them to give opening remarks at the event or to judge/announce the winners.

PHASE 4: THE COMPETITION

This phase is the only one that will take place in the venue. Bring a laptop for the welcome presentation. You may also wish to bring the following optional materials:

- Waters and snacks for participants
- Blank nametags and markers
- If the venue has multiple entrances, signs that point to the main entrance.

These materials will not be provided by MathWorks.

4.1. EVENT DAY SETUP

Show up to the venue an hour early to set up the event space for participants.

4.1.1 (OPTIONAL) WELCOME TABLE

- Select a table from the venue or bring your own folding table to serve as a welcome table. You can use this as an opportunity to remind participants to sign up for the event on Devpost, have them create nametags, answer questions, and/or give out anything you'd like participants to have.

4.1.2 SLIDE DECK SETUP

- Open the Welcome slide deck on a laptop, connect this laptop to the projection system available in the room, and turn the projection system on.
- Make the slide deck full screen, displaying the first slide.

4.1.3 CHAIRS, DESKS, OUTLETS

- There should be enough chairs for each participant.
- Chairs and desks should be relatively close to outlets so that participants can charge their laptops as they work.

- If the venue space is too big for this, surge protectors with a long cord may be used to increase access to outlets.
- Tape these cords to the ground to prevent safety hazards.

4.2 WELCOME PARTICIPANTS

4.2.1 WELCOME PRESENTATION

- 5 minutes after the scheduled start of the event, present the Welcome Presentation
- The Welcome Presentation should take approximately 15 minutes for most problem statements, followed by a brief Q&A.

4.3. PARTICIPANTS WORK ON PROJECTS

For most of the event, participants will work solo or in groups to create a solution to the problem statement. Be available to answer questions and engage with participants.

- Teams should use GitHub, MATLAB Drive, Google Drive, File Exchange, or some other online repository for sharing code and version control
- Each problem statement comes with some starter code or getting started resources, which can be found in the 'Resources for Participants' section of the problem statement. Share this somewhere accessible for participants, like on the Devpost page.

4.4. JUDGING

4.4.1. SUBMIT PROJECTS

- At least 45 minutes before the end of the event, each team should create and submit a project on Devpost. This asks for:
    - Team Name
    - Team Members
    - Repository URL to project
    - What programming languages were used in the submission
- The repository should be made public so the judges can access it
- Only one person per team should submit
    - The repository may not be modified after the deadline.

4.4.2 EVALUATE SUBMISSIONS

- Each judge should be responsible for judging 4-8 submissions.
- For each submission, judges will need to:
    - Verify that the repository was not modified after the submission deadline
    - Score the submission against the judging rubric, which can be found in the "Judging and Scoring" section of the problem statement.
    - Add the team's name, repository link, and project score to the scoring spreadsheet created previously.
- Each submission should take about 7 minutes to judge

4.4.3 DETERMINE WINNERS

- Sort the scoring spreadsheet, which was filled in during the previous step, in descending order by the 'Score' column.
- Once the list is sorted, the teams in row 2, 3, and 4 are the First, Second, and Third place winners, respectively.
- Fill out the winner's certificates with the names and/or team name of each participant on a winning team

4.4.4 ANNOUNCE WINNERS

- Gather all participants and announce the first, second, and third place winners.
- Distribute prizes and winner's certificates to each member of the winning teams.
- Thank all participants
- This concludes the in-person event, participants may leave at this time and you can start cleaning up the event space.

PHASE 5: FOLLOW-UP STEPS

This phase focuses on the follow up tasks **after** the in-person portion of the competition has concluded.

5.1 FOLLOW-UP COMMUNICATIONS

5.1.1 SEND THANK-YOU EMAIL

- You may choose to email all participants thanking them for their participation and congratulating the First, Second, and Third place winners.
- If you wish to use them, fill in and attach a digital copy of the 'Certificate of Participation', found in the certificate templates, to this email for all participants

5.1.2 TELL US HOW IT WENT!

- We would love to hear about your event! Feel free to send us an email at hackathon@mathworks.com letting us know how it went, how many people attended, if you have any feedback on this repository, and/or if there's anything else you want to share.
- If you make a post about your hackathon on Instagram, feel free to tag @matlab_students to share your event with our followers.

PROBLEM STATEMENTS

This section contains all approved Problem statements that can be used for MathWorks Hackathons, along with their corresponding starter code, judging rubric, and GitHub link.

1. FITNESS TRACKER

Suggested Resource Group: AI

1.1 PROBLEM STATEMENT

Fitness trackers are used to track fitness data as you live your everyday life. The technology used in these types of devices are low tech and can be recreated from home. In this challenge you will use MATLAB and MATLAB mobile to make your own fitness tracker.

Using sensor data retrieved from your phone, the goal is to create a model to turn this data into useable results to inform someone about how effective their workout was. This output data could include any sort of fitness data such as calories burned, steps taken, or flights climbed. Your task is not only to figure out what information you want to output, but also how to make a model to output this information. If possible, these models should utilize machine or deep learning techniques. Once you have a model and results, you will need to present your findings in an easy-to-understand manner.

This slide can be used in the 'Welcome Presentation' to introduce this topic:

- Problem Statement Slide for Welcome Presentation.pptx

1.2 RESOURCES FOR PARTICIPANTS

GitHub Link containing event-day resources: https://github.com/mathworks/matlab-mobile-fitness-tracker

1.3 JUDGING AND SCORING

We recommend asking participants to submit their code and a 5-minute or less presentation of the project. The presentation could be a video, a slideshow, a written explanation, etc. They should indicate which file(s) are the presentation in the submission.

Submissions for the Fitness Tracker problem statement should be evaluated as follows:

Fitness Tracker MATLAB Model	**Points (70)**
Creativity - Innovative, creative, and original work	10
Difficulty and Mastery - Level of MATLAB knowledge demonstrated in executing the tasks	10
Functionality - Error-free and runs without issues	10
Readability - Clean, organized, and easy to comprehend	10
Data Visualization - Clear and insightful graphics	10
Model Making - Transitioned model ideas into a viable model implementation	10
Advanced Model Making - Use of machine or deep learning techniques in model	10

Presenting Results - Report, Presentation or Video	**Points (30)**
Creativity - Interesting delivery methods, innovative and informative	10

Quality - Technical execution of material and attention to detail	10
Concept - Engaging, coherent, and appropriate	5
Clarity - Message is clear and well-communicated	5

Past submission and winner examples that judges can use for reference:

- https://matlab-hackathon-greece.devpost.com/project-gallery
- https://matlabthon-polimi.devpost.com/project-gallery

1.4 SUGGESTED PRE-WORK

We recommend participants complete the following pre-work before the event:

- MATLAB Onramp
- Machine Learning Onramp

If your audience is already familiar with these topics or if you plan to introduce these topics through a workshop instead, you may choose to skip these.

2. MICROGRID ON MARS

Suggested Resource Group: Electrification

2.1 PROBLEM STATEMENT

You are an engineer stationed on Mars and have a rather urgent design task ahead of you. A rogue meteor has rendered the original solar array inoperative - the microgrid system now has only battery power and time is running short. You must design a new microgrid system that will provide uninterrupted power over a seven sol period (1 sol = 24 hours, 37 minutes). The system must also keep voltage between 500V and 520V at all times, as various sensitive pieces of life-support equipment cannot tolerate voltage levels outside these bounds. You only have a few Earth hours to complete your design.

This slide can be used in the 'Welcome Presentation' to introduce this topic:

Problem Statement Slide for Welcome Presentation.pptx

2.2 RESOURCES FOR PARTICIPANTS

GitHub Link containing event-day resources: https://github.com/mathworks/microgrid-on-mars-activity

2.3 JUDGING AND SCORING

Participants should submit all of their completed Simulink files.

Solution Quality	Points (50)
Design Space Exploration	10
Maximum Power Transfer	10
Develop Maximum Power Point Tracking (MPPT) Control	10
Battery Voltage Control	10

Integrate and Test the Full System	10
Best Use of MATLAB & Presenting Quality	**Points (50)**
Creativity: Interesting delivery methods, innovative and creative solutions	15
MATLAB/Simulink Mastery: Knowledge and execution	25
Clarity: Message is clear and well-communicated	10

Complete, well-done examples that judges can use for reference:

- Complete

2.4 SUGGESTED PRE-WORK

We recommend participants complete the following pre-work before the event:

- MATLAB Onramp
- Simulink Onramp

If your audience is already familiar with these topics or if you plan to introduce these topics through a workshop instead, you may choose to skip these.

3. PREDICTING DIABETES IN PATIENTS

Suggested Resource Group: AI

3.1 PROBLEM STATEMENT

This challenge tasks participants to determine whether a patient admitted to an ICU has been diagnosed with a particular type of diabetes, Diabetes Mellitus, using data from the first 24 hours of intensive care. This would inform the best way to treat the patient. Participants must make their own decisions on how they want to split and process the data, train their models, and present their findings.

Participants will need to download the data directly from Kaggle.

This slide can be used in the 'Welcome Presentation' to introduce this topic:
Predicting Diabetes Problem Statement Slide.pptx

3.2 RESOURCES FOR PARTICIPANTS

- Participants must download the training dataset from Kaggle:
    - https://www.kaggle.com/competitions/widsdatathon2021/data
- Example code that tackles this problem:
    - https://github.com/mathworks/predicting-diabetes-mellitus

3.3 JUDGING AND SCORING

Each Team will create a 5-minute presentation (this could be a video, slide deck, essay, or other medium of their choice) along with their code. This should explain what their project does and how they made their decisions. Judges will evaluate these submissions based on the following rubric:

Solution Quality	Points (70)
Data Mastery: Makes informed choices about how they choose to clean, process, and separate their data.	20
Use of Machine Learning: Can justify their algorithm choices, iterates on their models, and can demonstrate that the model achieves some level of accuracy	20
Functionality: Error-free and runs without issues	10
MATLAB proficiency: Level of MATLAB knowledge demonstrated in executing the tasks	10
Readability: Clean, organized and easy to comprehend	10
Presenting Quality	**Points (30)**
Creativity: Interesting delivery methods that are informative	10
Quality: Technical execution of material and attention to detail	10
Concept: Engaging, coherent and appropriate	10

3.4 SUGGESTED PRE-WORK

We recommend participants complete the following pre-work before the event:

- MATLAB Onramp
- Machine Learning Onramp

If your audience is already familiar with these topics or if you plan to introduce these topics through a workshop instead, you may choose to skip these.

4. PREDICTING BUILDING ENERGY CONSUMPTION

Suggested Resource Group: AI

4.1 PROBLEM STATEMENT

This hackathon challenges participants to develop a machine learning model that predicts building energy consumption based on factors like weather, building characteristics, and more. With buildings being major contributors to global energy use and carbon emissions, accurate predictions could enable organizations to implement targeted strategies to reduce energy waste, such as retrofitting high-use buildings to be more energy efficient. This challenge offers participants the chance to see what it's like to work with a real dataset on a current global issue while enhancing their technical skills.

Participants must make their own decisions on how they want to split and process the data, train their models, and present their findings.

This slide can be used in the 'Welcome Presentation' to introduce this topic:
Predicting Building Energy Consumption Problem Statement Slide.pptx

4.2 RESOURCES FOR PARTICIPANTS

- Participants must download the training dataset from Kaggle:
    - https://www.kaggle.com/competitions/widsdatathon2022/data
- Example code that tackles this problem:

    - https://github.com/mathworks/predicting-building-energy-consumption

A4.3 JUDGING AND SCORING

Each Team will create a 5-minute presentation (this could be a video, slide deck, essay, or other medium of their choice) along with their code. This should explain what their project does and how they made their decisions. Judges will evaluate these submissions based on the following rubric:

Solution Quality	**Points (70)**
Data Mastery: Makes informed choices about how they choose to clean, process, and separate their data.	20
Use of Machine Learning: Can justify their algorithm choices, iterates on their models, and can demonstrate that the model achieves some level of accuracy	20
Functionality: Error-free and runs without issues	10
MATLAB proficiency: Level of MATLAB knowledge demonstrated in executing the tasks	10
Readability: Clean, organized and easy to comprehend	10
Presenting Quality	**Points (30)**
Creativity: Interesting delivery methods that are informative	10
Quality: Technical execution of material and attention to detail	10
Concept: Engaging, coherent and appropriate	10

4.4 SUGGESTED PRE-WORK

We recommend participants complete the following pre-work before the event:

- MATLAB Onramp
- Machine Learning Onramp

If your audience is already familiar with these topics or if you plan to introduce these topics through a workshop instead, you may choose to skip these.

5. DRONE SIMULATION

Suggested Resource Group: Robotics

5.1 PROBLEM STATEMENT

The goal of this challenge is to optimize a drone's ability to autonomously follow a path in a simulated, 3D environment.

In this project, participants are tasked with designing a robust path-planning algorithm for their drone. The simulation model provided is pre-configured with essential control systems and image processing blocks, which process the drone's camera feed to detect the track. Participants will use Simulink and Stateflow to design the decision-making logic for the drone's path planning, demonstrating a solid understanding of control system integration, state machine design, and real-time simulation testing.

5.2 ORGANIZER SET-UP

At the beginning of this event, instead of using the 'Welcome Presentation' template, someone should present the 'Intro Presentation', which introduces participants to drone simulation. This presentation should take about an hour.

- Intro Presentation.pptx

Additionally, you will have to decide how many paths to provide to participants, between 2 and 6 depending on the audience and how much time they will have. The first two paths are provided with the problem statement, but if you wish to provide additional tracks to evaluate the participants on, you may do so using the Track Builder, which is explained on slide 23 in the 'Intro Presentation' mentioned in the previous paragraph.

5.3 RESOURCES FOR PARTICIPANTS

Participants can access the environment for this challenge by installing the Simulink Support Package for Parrot Minidrone and typing the following command in the command window:

```
>>parrotMinidroneCompetitionStart
```

These instructions and other resources can be found here, and should be shared with participants at the beginning of the event:

- https://github.com/mathworks/drone-simulation

5.4 JUDGING AND SCORING

This is the rubric for this challenge:

Solution Quality	Points
Track 1	20
Track 2 (model **must** work on track 1)	20
Track 3 (if applicable) (model **must** work on track 1)	20
Track 4 (if applicable) (model **must** work on track 1)	20
Track 5 (if applicable) (model **must** work on track 1)	20

For each track, the score will be evaluated on the following:

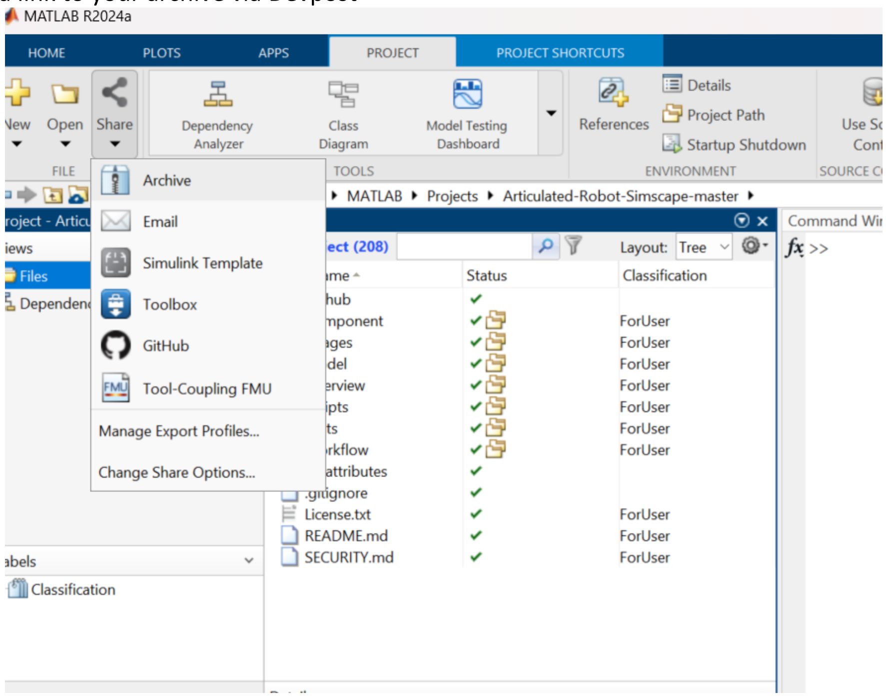
- Accuracy of interpreting the track
- Image processing formula
- Stateflow algorithm
- Simulation

Participants will submit their Simulink model, and judges should load the models, run the simulation, and evaluate based on the criteria above. Note: Please update the virtual world in track builder before running the simulation

Alternatively, you can ask students to create a presentation of their workflow and show the simulations. These can either be presented live, recorded, or just submitted as a slide deck. This is up to you, but explain these expectations to participants on Devpost and at the beginning of the event.

Instructions given to the students on how to share the model:

- After finalizing the Simulink model, export your model as a project zip file (Project->Share->Archive)
- Save as Student Name and ID (Studentname_ID.prj)
- Upload this archive to an online storage system that allows sharing, such as google drive, GitHub, etc.
- Submit a link to your archive via Devpost



5.5 SUGGESTED PRE-WORK

We recommend participants complete the following pre-work before the event:

- [Simulink Onramp]
- [Stateflow Onramp]

If your audience is already familiar with these topics or if you plan to introduce these topics through a workshop instead, you may choose to skip these.

5.6 ENVIRONMENT REQUIREMENTS

This project requires an installation of MATLAB® Desktop and **does not work on MATLAB® Online™.**

The following products are also required:

- Simulink®
- Simulink® 3D Animation™
- Stateflow®
- Aerospace Toolbox
- Computer Vision Toolbox™
- Control System Toolbox™
- Signal Processing Toolbox™
- Embedded Coder®

Not all operating systems are compatible with the Simulink Support Package for Parrot Minidrones, which is used for this project. Check platform compatibility here

6. SIGNAL DECRYPTION CHALLENGE

Suggested Resource Group: **Custom**. Some helpful references if you choose to provide some:

- Signal Analysis and Visualization – Learn how to use the Signal Analyzer app
- Practical Introduction to Time-Frequency Analysis – Understand what DTMF signaling is and how to analyze DTMF signals in time and frequency
- DFT Estimation with the Goertzel Algorithm – Efficiently detect DTMF tones using the Goertzel algorithm
- Design Peak and Notch Filters – Learn how to design filters to isolate or remove specific frequencies
- DTMF Generator and Receiver – Explore Simulink models for generating and decoding DTMF signals

6.1 PROBLEM STATEMENT

This challenge is designed to help you build core signal processing skills while experiencing the excitement of decrypting hidden audio messages. You will work with Dual-Tone Multi-Frequency (DTMF) signals—commonly used in telephone systems—to decode passwords and uncover secret messages. Participants can choose to start with the five optional experiments to hone their skills, then are challenged with decoding a password from an audio file, which will allow them to unlock a .p file where they will decode the final secret message.

This slide can be used in the 'Welcome Presentation' to introduce this topic:

- Problem Statement Slide for Welcome Presentation.pptx

6.2 RESOURCES FOR PARTICIPANTS

GitHub Link containing event-day resources: https://github.com/robert-dobre/Signal-Decryption-Challenge

6.3 JUDGING AND SCORING

Attendees not only need to submit their results but also are encouraged to submit a presentation to explain how they solved the problem. The presentation could be a video, a slideshow, a written explanation, etc. They should indicate which file(s) are the presentation in the submission.

Total: 100 points + 10 bonus points

Evaluation Criteria	Description	Points
Data Import, Analysis & Visualization	Successfully import audio data and use appropriate visualizations (e.g., time-domain plots, spectrograms, DTMF tone detection) to support analysis.	20
Password Decryption	Accurately decode the 11-character password from the DTMF audio signal using signal processing techniques.	30
Secret Message Decryption	Successfully unlock and extract the hidden message from the secret audio file using filtering or other signal processing methods.	40
Presentation Quality	Deliver a clear, well-structured, and engaging presentation (in-person or recorded) that demonstrates understanding of the problem, methodology, and results.	10
Scalability (Bonus)	Develop a scalable solution using MATLAB functions or a Simulink model that can automatically decode passwords from DTMF audio inputs.	+10

Notes:

- The correct answers can be found here: Passwords and Secret Message
- The secret message does not need to be decoded exactly the same as the correct one. A message pronounced similarly could also be fine.
- A hint could be offered after a certain amount of time, especially if teams are not progressing as expected, for instance, referring to the examples in the resources.

6.4 SUGGESTED PRE-WORK

We recommend participants complete the following pre-work before the event:

- MATLAB Onramp
- Signal Processing Onramp

If your audience is already familiar with these topics or if you plan to introduce these topics through a workshop instead, you may choose to skip these.

7. DATASTORM CHALLENGE

Suggested Resource Group: **Custom**. Some helpful references if you choose to provide some:

- Matrices and Arrays
- Data Exploration documentation
- Descriptive Statistics and Insights

7.1 PROBLEM STATEMENT

Meet Memby, a young data scientist passionate about uncovering hidden patterns in nature. Participants have a mission to help Memby with specific data analysis tasks to uncover hidden patterns in Mathtown's weather data.

The DataStorm problem statement is great for shorter events with participants that are new to MATLAB. It provides them with a weather dataset and challenges them with 10 data extraction and analysis tasks that solve real-world problems. We suggest you familiarize yourself with the problems and potential solutions ahead of time, as participants may have questions.

We recommend the following time allotment for this event:

- 1-2 hours for an audience that may have experience with coding or MATLAB
- 2-3 hours for participants that are brand new to coding and MATLAB.

This slide can be used in the 'Welcome Presentation' to introduce this topic:

- DataStorm Slide for Welcome Presentation

7.2 SET UP

This entire hackathon is in MATLAB Grader, so you will need to set up the Course that participants will complete during the event. You will need to reach out to hackathon@mathworks.com to gain access to the course materials, so we recommend reaching out as soon as you know you want to run this problem statement.

Detailed information about the problem statement and set-up instructions can be found here:

- DataStorm Instructions for Organizers

7.3 RESOURCES FOR PARTICIPANTS

Since this hackathon is in MATLAB Grader, participants will need to have a MathWorks account and have opened MATLAB Grader in advance of the hackathon. Include the following instructions on the Devpost page:

1. Create a MathWorks account at https://www.mathworks.com/mwaccount/ ahead of time.
2. Go to https://grader.mathworks.com/ and check that your login works here.

7.4 JUDGING AND SCORING

MATLAB Grader will automatically grade each problem that the participants submit. Follow the instructions in the Instructions for Organizers document to export each participants' scores and rank participants based on how many problems they solved. This is done using the following MATLAB Live Script:

- datastormAutoGrading

It's important to let participants know that ties will be broken based on the number of individual tests passed, then by the overall size of their submissions.

7.5 SUGGESTED PRE-WORK

Participants do not need to complete any pre-work before this event, though you may choose to offer the following:

- MATLAB Onramp
- Get familiar with MATLAB Documentation

APPENDIX AND ATTACHMENTS

This Appendix contains all additional files you may need.

DEVPOST GUIDE

- Code of Conduct

HACKATHON INTRODUCTION SLIDE DECK

- Welcome Presentation

CERTIFICATE TEMPLATES

- Hackathon Certificates

PROMOTIONS PLAYBOOK

- Hackathon Promotional Playbook

- Social Media Templates
- Poster Templates

COMMUNICATIONS TEMPLATES

- Registration Email